The Michigan Model for Diabetes – Chronic Kidney Disease User Manual

Version 1.0
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Produced by the University of Michigan
Michigan Diabetes Modeling Group
https://michigandiabetesmodelinggroup.github.io/

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List of Abbreviations

HbA1c	Glycated hemoglobin
BMI	Body mass index
CAD	Coronary artery disease
CVD	Cardiovascular disease
MI	Myocardial infarction
CHD	Coronary heart disease
CHF	Congestive heart failure
DR	Diabetic retinopathy
MMD-CKD	Michigan Model for Diabetes-Chronic Kidney Disease
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
eGFR	Estimated Glomerular Filtration Rate
UPCR	Urine protein to creatinine ratio
UACR	Urine albumin to creatinine ratio
PTCA	Percutaneous transluminal coronary angioplasty
CABG	Coronary artery bypass graft
ACE-I	Angiotensin converting enzyme-inhibitor
ARB	Angiotensin receptor blocker
QALE	Quality-adjusted life expectancy
QALYs	Quality-adjusted life years
IEST	Indirect Estimation and Simulation Tool

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1. Introduction and Background

The Michigan Model for Diabetes-Chronic Kidney Disease (MMD-CKD) is a computerized disease model that enables the users to simulate the progression of chronic kidney disease, coronary heart disease, and cerebrovascular disease overtime in type 2 diabetes (T2DM) patients with chronic kidney disease. Transition probabilities can be a function of individual characteristics, current disease states or treatment status. The model also estimates the medical costs of diabetes and the three above comorbidities and complications, as well as the quality of life related to the current health state of the subject.

Transition probabilities and other model parameters implemented in the MMD-CKD were obtained by both analyzing individual level secondary data and synthesizing the published literature. The main source of individual level data is the Chronic Renal Insufficiency Cohort (CRIC) Study. We also used individual level data from the Action to Control Cardiovascular Risk in Diabetes (ACCORD) study for developing part of the model.

Our model allows a user to control risk factor changes by defining treatment thresholds and compliance rates for chronic kidney disease, hyperglycemia, dyslipidemia, and hypertension, and compliance to quitting smoking, taking aspirin and anticoagulants. Given the fact that modern medicines have largely decreased the complication rate in type 2 diabetes through management of these risk factors, it is important to explicitly model these management strategies and allow users to modify them to match the specific scenarios that they are simulating.

Most of the risk equations used in the MMD-CKD were developed using CRIC study data. The CRIC Study included a racially and ethnically diverse group of adults with a broad spectrum of renal disease severity, half of whom will have diagnosed diabetes mellitus, the remainder having a broad array of nondiabetic renal disease. Patients aged 21 to 74 yr with chronic renal insufficiency (CRI) were enrolled in the CRIC Study. Age-related entry criteria for GFR were used to limit the proportion of older individuals recruited with age-related diminutions of GFR (GFR [ml/min per 1.73 m^2]), but otherwise nonprogressive CRI: 20-70 for age 21 to 44 yr, 20-60 for age 45 to 64 yr, and 20-50 for age 65 yo 74. The estimated GFR to define eligibility is based on the simplified MDRD estimating equation. Among the 1441 CRIC subjects with T2DM included our analyses for model development, 46%, 31%, 18%, and 4% are non-Hispanic Black, non-Hispanic White, Hispanic, and other races, respectively. In light of this, and recognizing that the other data sources for our model are studies that were conducted in the United States and Western Europe, and considering the difference in medical practice across countries, caution should be applied when model results are extrapolated to populations that differ significantly from the model target population: relatively young (20-81 years of age) non-Hispanic Black, non-Hispanic White, and Hispanic populations with type 2 diabetes in the United States and Western Europe.

Despite this, the R Shiny Web App, which houses our model, allows users to adjust parameters to better suit their own situations. In addition, we share the R code with users who need to make additional changes. For example, when applying the model to a population in a country with less access to revascularization procedures, users can adjust the transition probabilities to match the revascularization procedure rates in their countries.

2. About the RShiny Web App

We use an RShiny Web App to host and distribute the MMD-CKD 1.0. This Web App provides a self-explanatory user interface and allow use to input study population, define simulation length, number of iterations, modify cost, quality of life related parameters, and treatment related parameters with ease. It also provides summary statistics in both figure and table formats.

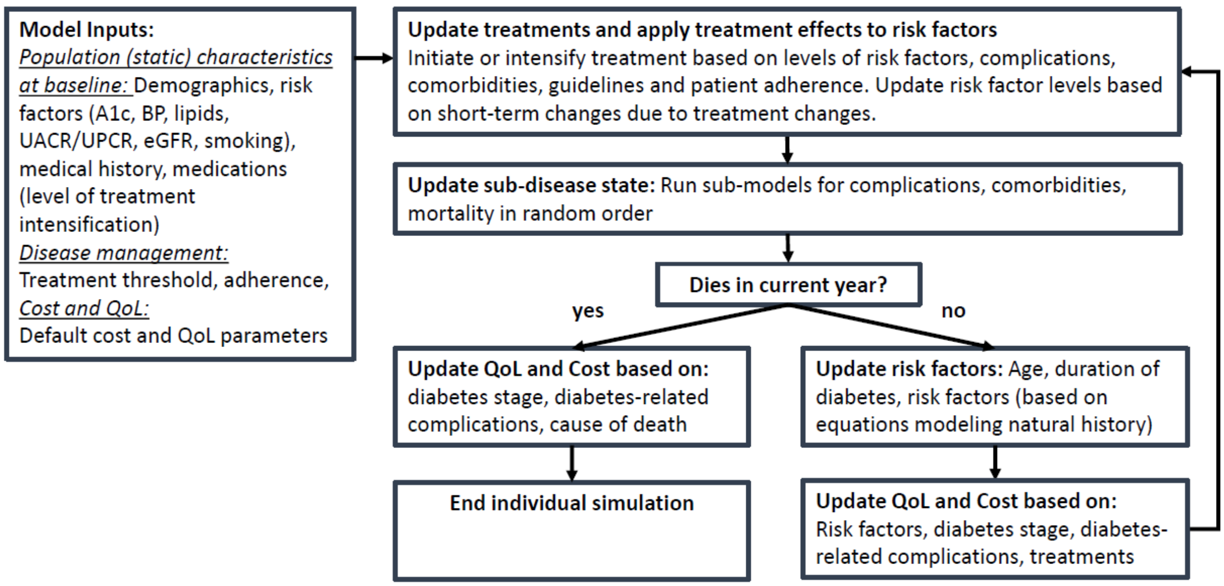
It also provides raw simulated data for all simulated individuals, e.g. risk factors, complications status, yearly medical cost and utility score for each simulated year. Using the raw results, users can also write their own programs to summarize other quantities of their own interest.

This RShiny Web App will also evolve over time to allow more functions, such as one-way sensitivity analysis, probability sensitivity analysis, handling missing data in the individual-level baseline population data.

3. MMD-CKD Modules and Its implementation

For each subject, the model software reads in or simulates the subject's baseline characteristics (See **Section 5.1** for details) and then advances the subject through a specific number of years or until death. Each year, the model updates in the four stages as indicated by blue blocks in the following figure 1, including:

Figure 1. MMD-CKD Simulation flow chart and algorithm



1) Update treatments based on patient's risk factor levels and medical history, treatment guidelines, and patient's adherence parameters. The program then calculated the short-term changes in risk factors changes due to treatment changes. See **Section 4.4** for details. This step is skip for the first simulation cycle.
2) Update disease states and complications based on transition probabilities which can be functions of individual characteristics, current disease states or treatment status. See **Section 4.2** for details of model specification. Hypoglycemia events are also simulated based on parameters defined in **Section 4.3.**
3) Update risk factors (i.e. weight/BMI, HbA1c, fasting glucose, systolic blood pressure (SBP)/diastolic blood pressure (DBP), urine albumin creatinine ratio (UACR)/urine protein albumin creatinine ratio (UPCR), estimated Glomerular Filtration Rate (eGFR), lipids) according natural history of changes or medication modified natural history changes from beginning of the year to the end of the year. See **Section 4.5** for details.
4) Calculate cost and utility values for the specific year according to risk factors, medications, and history of complications and comorbidities. See **Section 4.6 and 4.7** for details.

4. MMD-CKD Structures and Parameters

4.1. Population Input Module

Populations can either be inputted as data (to be used in a simulation), or set by specifying a distribution (to be used in Estimation or for randomly generating population sets). It is the responsibility of the users of MMD-CKD to ensure that only valid values are entered as the software applies a few data entry checks.

4.1.1. Individual level data

If you wish to use individual-level data, the variables needed for each subject are listed in the following table.

Table 1. Variable specificiations for individual-level data input

Varaible Name	**Definition**	**Legal Range**	**Notes**
Demographics Characteristics			
Age	Current age in years	[1,100]	
Duration_Of_Diabetes	Duration in years since diagnosis of diabetes	<=Age	
Male	Gender Variable	1=Male, 0=Female	
Race	Race	Non-Hispanic White Non-Hispanic Native American Non-Hispanic Black/African American Non-Hispanic Asian Hispanic	
Weight	Weight in kilograms[1.0kg = 2.2 pounds]	[30, 400]	
Height	Height in meters [1.0meter=39 inches]	[0, 2.5]	
Current Risk Factors			
SBP	Systolic blood pressure(mmHg)	[70, 220]	
DBP	Diastolic blood pressure(mmHg)	[20, 120]	
Smoke	Smoking Status	0 = Non-smoker, 1=Smoker	
HDLCholesterol	High-density lipoprotein cholesterol in mmol/L [1mmol/L = 38.6mg/dl]	[0.3, 5]	

LDLCholesterol	Low-density lipoprotein cholesterol in mmol/L [1mmol/L = 38.6mg/dl]	[0.3, 11]	
Triglycerides	Triglycerides in mmol/L [1mmol/L = 88.5mg/dl]	[0, 20]	
HbA1c	Hemoglobin A1c(%)	[0, 20]	
eGFR	Estimated Glomerular Filtration Rate (mL/min/1.73m^2)	(0, 120]	
UPCR_gperg	Urine protein to creatinine ratio (g/g)	(0,30]	
UACR_gperg	Urine albumin to creatinine ratio (g/g)	(0,30]	
Afib	Atrial fibrillation	1=Yes, 0=No	
Disease Status (Within each sub-model defined below, one and only one variable should be set to 1)			
Cerebrovascular disease sub-model			At most, only one of them can be set to 1.
No_Cerebrovascular_Disease	No cerebrovascular disease	1=Yes, 0=No	
Hx_Is_MinorDisability	History of Ischemic Stroke with minor/no disability	1=Yes, 0=No	
Hx_Is_MajorDisability	History of Ischemic Stroke with major disability	1=Yes, 0=No	
Hx_Hs	History of Hemorrhagic Stroke	1=Yes, 0=No	
Coronary heart disease sub-model			
No_CVD	No history of coronary heart disease	1=Yes, 0=No	One and only one of these categories should be set to 1.
CADwoMI	Coronary artery disease without history of MI or heart failure	1=Yes, 0=No	
CHFwoMI	History of heart failure but not MI	1=Yes, 0=No	
CADwProc	History of revascularization procedure with no history of MI	1=Yes, 0=No	
Survive_MI	History of MI(can be more than once) with no history of heart failure	1=Yes, 0=No	
CHF_after_MI	History of heart failure and history of MI	1=Yes, 0=No	
Nephropathy sub-model			
No_ESKD	No nephropathy		One and only one of these categories should be set to 1.
ESKD_Dialysis	End stage renal disease with need of dialysis but no history of transplant	1=Yes, 0=No	
ESKD_Transplant	End stage renal disease with history of transplant	1=Yes, 0=No	
Medication			
IntensiveLifeStyle	Diet and exercise	1=Yes, 0=No	

Metformin	Metformin	1=Yes, 0=No	There are five stages for anti-hyperglycemia treatment in MMD. These five stages are mutually exclusive of each other. At most, only one of them can be set to 1. [a]
OtherOralMedication	Two or more oral/non-insulin medication (e.g., metformin+sulfonylureas)	1=Yes, 0=No	
BasalInsulin	Basal Insulin	1=Yes, 0=No	
Insulin	Intensive bolus insulin	1=Yes, 0=No	
Beta_Blocker	Whether a subject is taking beta-blocker	1=Yes, 0=No	
ACE_ARB	Whether a subject is taking ACE/ARB	1=Yes, 0=No	
Number_of_BPMeds	Number of hypertension drug(exclude beta-blocker) the subject is taking	[0, 100]	
Statin	Whether a subject is taking any medication for dyslipidemia	1=Yes, 0=No	
Aspirin/Clopidogrel	Whether a subject is taking aspirin	1=Yes, 0=No	
Warfarin	Whether a subject is taking warfarin	1=Yes, 0=No	
NOAC	Whether a subject is taking noval oral anti-coagulants	1=Yes, 0=No	
Sampling weights			
Sampling_Weight	Sampling weights assigned to each subject		If there is no sampling weights, enter 1 for each subject.

[a]Additional instructions to set up five variables of medications for anti-hyperglycemia treatment: 1) If a subject is on insulin therapy in which only basal insulin or only premixed insulin is used, s/he should be considered at the 4th stage treatment for hyperglycemia, and therefore only the variable BasalInsulin is set to be 1. 2) If a subject is on insulin therapy in which any of rapid-acting insulin, short-acting insulin, or intermediate-acting insulin is used, s/he should be considered at the 5th stage treatment for hyperglycemia, and therefore only the variable Insulin is set to be 1.

4.1.2 Specify a distribution

An alternative to inputting a data set with individual information is to simulate a baseline population using population level summary statistics. The RShiny Web App provides an input template that can be modified.

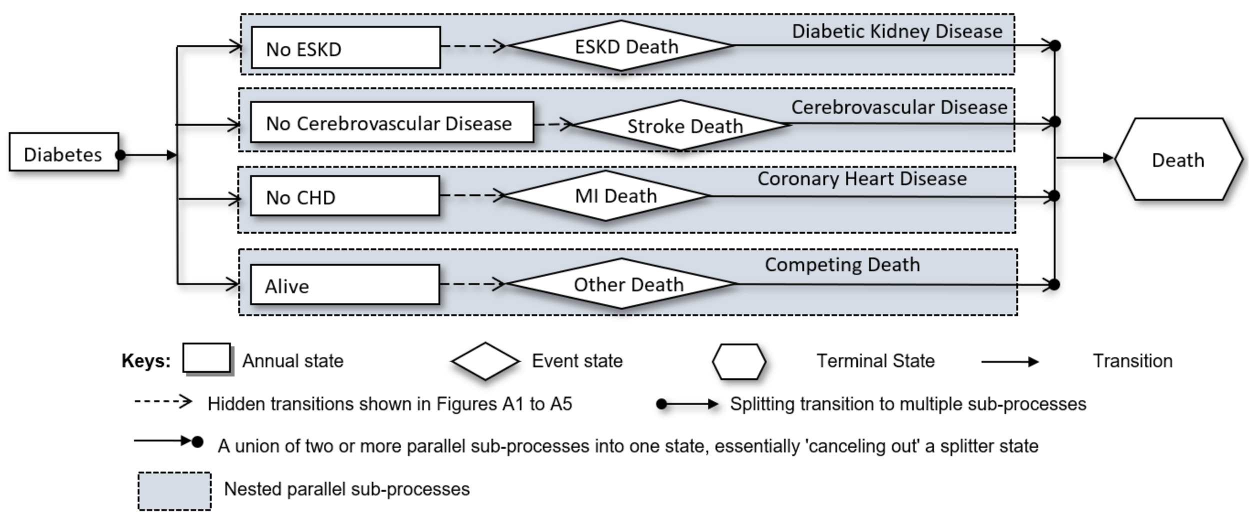
4.2 Disease Progression Module

4.2.1 Disease Progression Model Overall Structure

MMD-CKD is a discrete-state, discrete-time micro-simulation model in which the health status of each simulated subject is updated yearly. There are two types of states in the model: annual states and event states. Patients may stay in an annual state for one or more simulation cycles. Patients progress through event states, such as stroke and MI, instantaneously and transit to other annual states.

Figure 3 shows the overall structure of MMD-CKD model, which includes four parallel sub-models (cardiovascular disease [CVD], diabetic kidney disease, coronary heart disease [CHD] (Ye et al., 2015), and competing death).

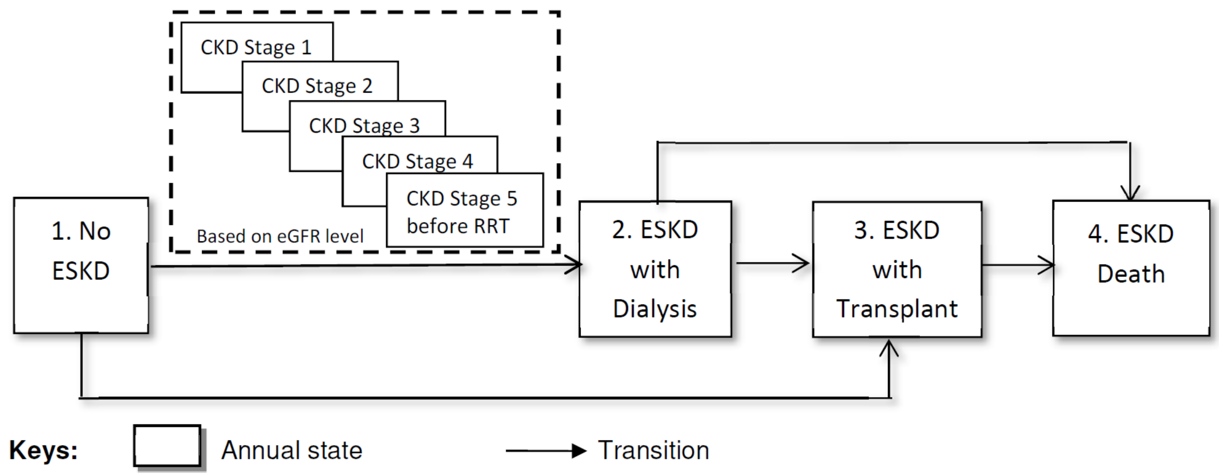
Figure 3. Overall Structure of MMD-CKD



ESKD: end stage kidney disease; MI: myocardial infarction; CHD: coronary heart disease

4.2.2 Structure and transition probabilities for nephropathy sub-model

Figure 4. Structure of kidney disease sub-model



ESKD: end stage kidney disease; RRT: renal replacement therapy

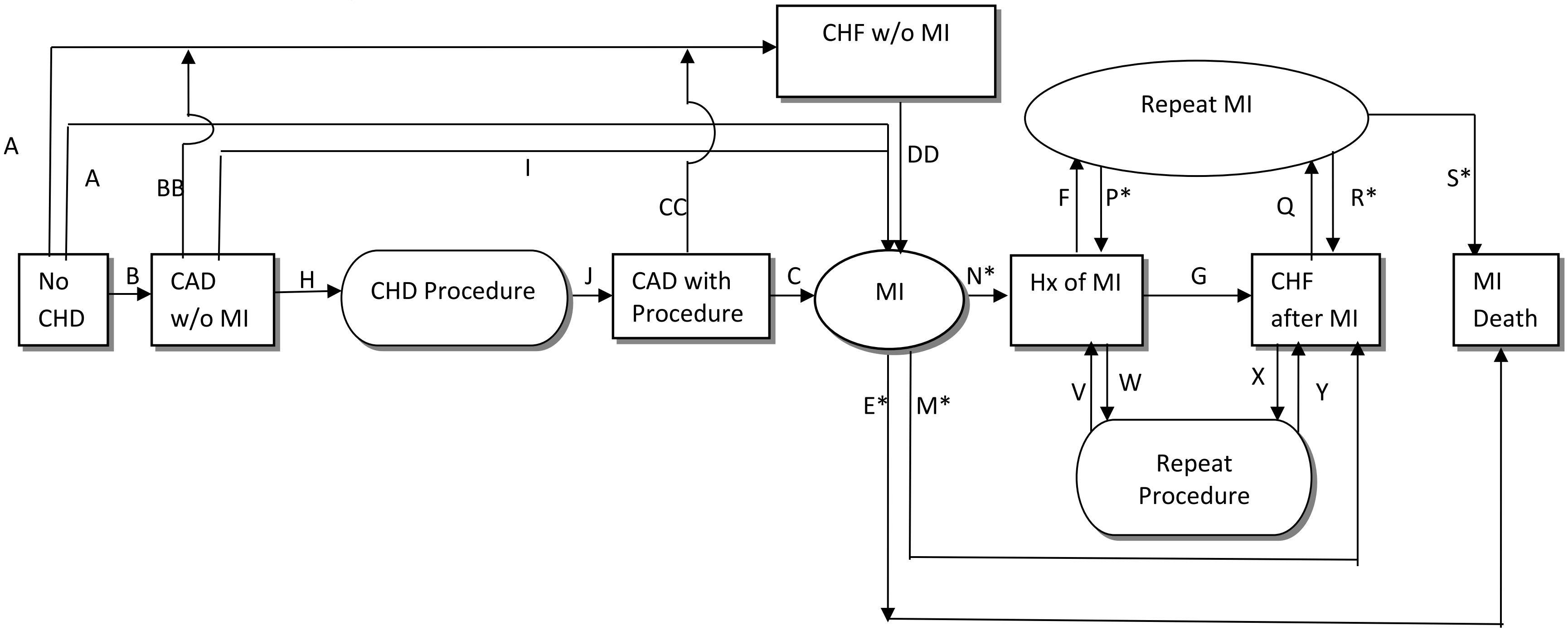
Table 2. Transition probability in nephropathy sub-model (Figure 4)		
Transition	Transition probability	Comments
1 to 2	Probability calculated based on equation derived from CRIC data analysis x (1-0.0806)	See Table 10
1 to 3	Probability calculated based on equation derived from CRIC data analysis x 0.0806	See Table 10
2 to 3	0.006 to 0.084 depends on age, gender, and race,	https://adr.usrds.org/2020/chronic-kidney-disease/6-healthcare-expenditures-for-persons-with-ckd This data of the renal transplant rates in dialysis patients in year 2013 was provided by KECC at the University of Michigan. The data was processed using the following criteria: 1) only the data for diabetes as ESRD cause was selected; 2) the data depended on age, gender, and race; 3) the data for White and Black was selected; 4) the data was divided by 100 to represent the yearly transition probability; and 5) the case counts for 0-21 age groups were probably too low to report the rates appropriately, and thus the transplant rates in 22-44 age groups were used for 0-21 age groups. See Table A2.a
2 to 4	0.0625 to 0.5344 depends on gender, age, race, hypertension (adjusted by other death causes)	https://usrds.org/media/1708/esrd_ref_h_mortality_2018.xlsx Table H.2_adj The data from the USRDS table was processed using the following criteria: 1) only the data for diabetes was selected; 2) the data depended on age, gender, and race; 3) the data was divided by 1,000 to represent the yearly transition probability; 4) based on average between 2011- 2016. See Table A2.b
3 to 4	0.0083 to 0.2487 depends on gender, age, race, hypertension (adjusted by other death causes)	https://usrds.org/media/1708/esrd_ref_h_mortality_2018.xlsx Table H.10_adj The data from the USRDS table was processed using the following criteria: 1) only the data for diabetes was selected; 2) the data depended on age, gender, and race; 3) the data was divided by 1,000 to represent the yearly transition probability; 4) based on average between 2011- 2016. See Table A2.c

Table 3. Rate of transition from "ESKD with dialysis" to "ESKD with transplant" per 1000 subjects (Figure 4)										
Age	Male					Female				
0-21	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic
22-44	8.4	4.3	4.3	4.3	4.3	8.3	3.5	3.5	3.5	3.5
45-64	4	2.6	2.6	2.6	2.6	2.8	1.9	1.9	1.9	1.9
65+	1.6	1.2	1.2	1.2	1.2	0.7	0.6	0.6	0.6	0.6

Table 4. Transition probability of "ESKD with dialysis" to "ESKD death" per 1000 patient years (Figure 4)										
	Male					Female				
	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic
30-39	122.6	78.8	85.6	58.0	65.4	140.5	96.1	95.1	62.5	75.8
40-49	145.7	89.9	96.6	78.9	81.1	158.5	104.6	102.3	81.9	89.5
50-59	184.1	117.6	121.9	100.4	105.6	188.8	125.7	121.3	101.7	109.9
60-64	225.4	145.0	159.6	129.0	137.2	217.7	146.2	156.5	123.0	134.7
65-69	266.3	173.7	191.2	154.1	169.9	250.9	169.9	179.6	143.3	162.6
70-74	314.2	206.1	230.2	188.6	208.0	294.9	203.7	209.7	171.1	198.3
75-79	365.7	243.7	267.0	221.4	254.1	336.5	239.5	238.9	202.8	237.4
80-84	444.6	297.5	332.3	286.0	327.1	403.0	285.5	296.6	250.1	299.7
85+	534.4	372.7	358.7	363.0	417.7	492.1	360.0	321.8	321.4	388.0

Table 5. Transition probability of "ESKD with transplant" to "ESKD death" per 1000 patient years (Figure 4)										
	Male					Female				
DIABETES	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic	Non-Hispanic White	Non-Hispanic Black/African American	Non-Hispanic Native American	Non-Hispanic Asian	Hispanic
30-39	14.8	15.1	14.0	8.3	9.4	15.5	14.4	15.5	8.4	9.7
40-49	21.8	21.7	24.0	12.6	15.3	23.2	20.7	25.2	12.8	16.1
50-59	39.4	36.9	38.5	23.8	27.9	38.5	32.9	39.0	22.5	27.2
60-64	61.4	55.3	67.9	38.9	45.0	60.3	50.4	66.1	37.2	44.1
65-69	81.1	72.0	80.0	52.2	65.7	79.8	65.2	79.9	50.2	64.2
70-74	107.2	91.0	103.8	73.8	87.5	110.9	84.3	109.4	73.8	89.7
75-79	146.7	125.1	168.8	105.5	132.6	153.9	117.4	173.5	107.5	138.6
80-84	212.8	157.6	178.2	147.1	176.5	204.9	137.1	222.2	136.5	169.5
85+	248.7	176.4	213.1	166.6	201.2	246.2	157.4	265.6	167.9	227.8

4.2.3 Structure and transition probabilities for CHD sub-model



Keys: Regular State Event State Module Transition

Figure 5. Coronary heart disease states and progression. CHD=coronary heart disease, CAD=coronary artery disease, CHF w/o MI=congestive heart failure without MI, MI=myocardial Infarction, CHF after MI=congestive heart failure after experience of MI, Hx=history, w/o=without, CHD procedure=revascularization procedure.

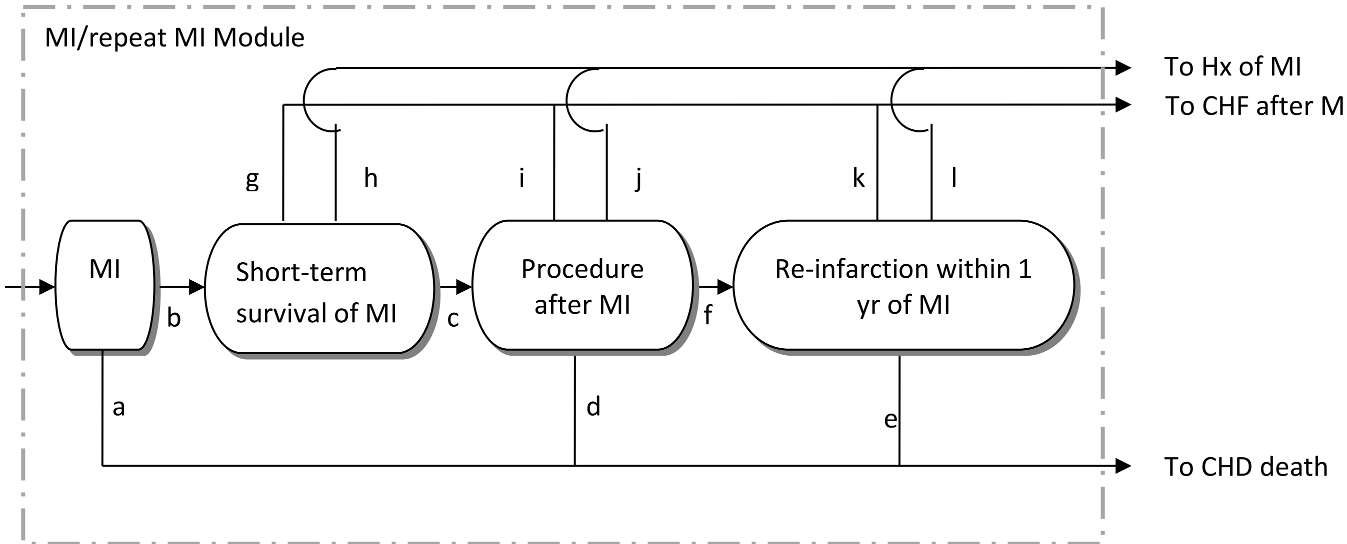


Figure 6. Myocardial infarction module. Ovals indicate instant states.

Table 6. Calibration and references for transition probabilities in the main CHD sub-model (Figure 5).		
Transition	Transition Probability	Reference
A (No CHD → MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 1.40448 . See Table 10.	Clarke et al.(2004); Avogaro et al (2007)
B (No CHD → CAD w/o MI)	UKPDS IHD hazard equation adjusted for direct medication benefit and by additionally a factor 6.27 .	
AA (No CHD → CHF w/o MI)	CHF hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 1.2834. See Table 10.	Fried LP et al. (1991)
I (CAD w/o MI → MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 3.370752 . See Table 10.	Clarke et al.(2004); Colhoun et al. (2004)
H (CAD w/o MI → CHD procedure)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 15.9258 . See Table 10.	
BB (CAD w/o MI → CHF w/o MI)	CHF hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 1.2834 . See Table 10.	Fried LP el al. (1991)
C (CAD with procedure → MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 2.782877 . See Table 10.	Clarke et al.(2004); Chaitman et al. (2009)
CC (CAD with procedure → CHF w/o MI)	CHF hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 1.2834 See Table 10.	Fried LP el al. (1991)
DD (CHF w/o MI → MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 0.140448 . See Table 10.	Clarke et al.(2004); Deedwania (2011); Mellbin et al (2011)
E* (MI → MI death)	See details in the MI/repeat MI module (Table 7 and Figure 6)	
M*(MI → CHF after MI)		
N* (MI → Hx of MI)		
F (Hx of MI → Repeat MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 2.501981. See Table 10.	Clarke et al.(2004); Mellbin et al. (2011); Jensen et al. (2011)
W (Hx of MI→ Repeat procedure)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by additionally a factor 6.42466 . See Table 10.	
G (Hx of MI → CHF after MI)	CHF hazard equation derived from CRIC study adjusted for direct medication benefit and by additionally a factor 1.2834 . See Table 10.	Fried LP el al. (1991)
P* (Repeat MI → Hx of MI)	See details in the MI/repeat MI module (Table 7 and Figure 6)	

Table 6. Calibration and references for transition probabilities in the main CHD sub-model (Figure 5).		
R* (Repeat MI → CHF after MI)		
S* (Repeat MI → MI death)		
Q (CHF after MI→ Repeat MI)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 2.182963 .	Clarke et al.(2004); Deedwania et al. (2011); Mellbin et al. (2011)
X (CHF after MI→ Repeat procedure)	MI hazard equation derived from CRIC study adjusted for direct medication benefit and by a factor 12.96009	
V (Repeat procedure → Hx of MI)	100% if subject does not have CHF 0% if subject have CHF	Cole et al. (2002)
Y (Repeat procedure → CHF)	100% if subject have CHF 0% if subject does not have CHF	

§Medications in this table refer to aspirin, lipid drug, ACE-inhibitor, and beta-blocker.

Table 7. Calibration and references for transition probabilities in MI/repeat MI module (Figure 6)

Transition	Transition Probability	Reference
a (MI → CHD death: fatal MI)	MI: Modified the UKDPS fatality equation by add gender effect. The new odds of death is -3.251+2.772*Ln(Age/52.59)+(HbA1c-7.09)*0.114+2.640+Female*Ln(3.5) We then calculate the probability of death using the odds and adjusted by a factor 0.18, disregard whether a patient has CHF or not.	Clarke et al.(2004); Colhoun et al. (2004); Roffi et al. (2013)
	Repeat MI: For subjects with CHF: Using the probability from the modified odds as described above. For subjects without CHF: Using the probability from the modified odds further adjusted by a factor 0.53	Clarke et al.(2004); Jensen et al. (2011)
b (MI → Short-term survival of MI)	1-transition probability in a	Clarke et al.(2004); Colhoun et al. (2004); Roffi et al. (2013)
c (Short-term survival of MI → Procedure after MI)	MI: 75%	Franklin et al. (2004); Jensen et al. (2011) Deedwania (2011)
	Repeat MI: 63%	
g (Short-term survival of MI → CHF after MI)	MI: For subject who has CHF before MI: 25% For subject who does not have CHF before MI: 25%×P(CHF)†	
	Repeat MI: For subject who has CHF before MI: 37% For subject who does not have CHF before MI: 37%×P(CHF)†	
h (Short-term survival of MI → Hx of MI)	MI: For subject who has CHF before MI: 0% For subject who does not have CHF before MI: 25%×(1-P(CHF))†	
	Repeat MI: For subject who has CHF before MI: 0% For subject who does not have CHF before MI: 37%×(1-P(CHF))†	
d (Procedure after MI → MI death)	MI: 12.5%	
	Repeat MI: 10%	
	MI: 8.75%	

Table 7. Calibration and references for transition probabilities in MI/repeat MI module (Figure 6)

f (Procedure after MI →Re-infarction within a year of MI)	Repeat MI: 9%	
i (Procedure after MI → CHF after MI)	MI: For subject has CHF before MI: 78.75% For subject has no CHF before MI: 78.75%×P(CHF)† Repeat MI: For subject has CHF before repeat MI:81%×P(CHF)† For subject has no CHF before repeat MI: 81%%×P(CHF)†	
j (Procedure after MI → Hx of MI)	MI: For subject has CHF before MI: 0 For subject has no CHF before MI: 78.75%×(1-P(CHF))† Repeat MI For subject has CHF before repeat MI: 0 For subject has no CHF before repeat MI: 78.75%×(1-P(CHF))†	Franklin et al. (2004); Jensen et al. (2011)
e (Re-infarction within a year of MI → MI death	17%	
k (Re-infarction within a year of MI → CHF after MI)	83%×P(CHF)†	
l (Re-infarction within a year of MI → Hx of MI)	83%×(1-P(CHF))†	

†P(CHF)=0.13*Age_Modifier*Gender_Modifier*0.45*Medication_Modifier for MI module; P(CHF)=0.13*Age_Modifier*Gender_Modifier Medication_Modifier for repeat MI module. For example, for a 60 years old male subject not on beta-blocker or ACE-Inhibitor, P(CHF) for the MI module = 0.13*0.87*0.86*0.45. Age and Gender modifier is described in Table A5. Medication_Modifier is as described Table 16.

Table 8. Age and Gender Modifier in Table 7 (Franklin et al., 2004)

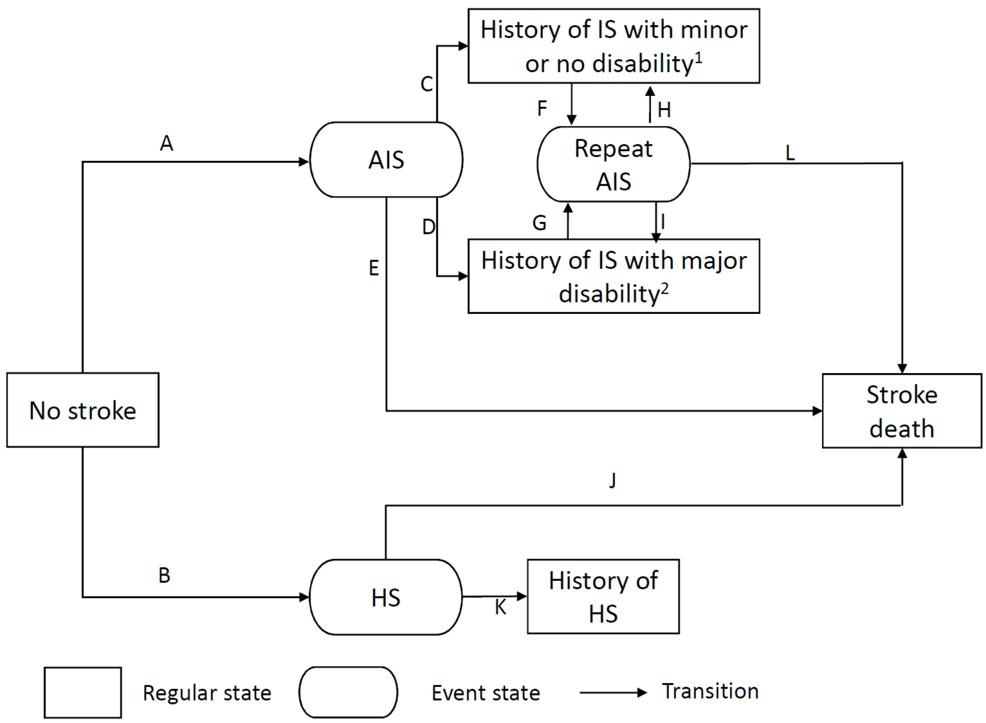
Factor	Category	Modifier
Age	<55	0.53
	55-64	0.87
	65-74	1.09
	>=75	1.51
Gender	Male	0.86
	Female	1.14

For example, for a 60 years old male subject not on beta-blocker or ACE-I, P(CHF) for the MI module = 0.13*0.87*0.86*0.45
Medication_Modifier is as Table 16.

4.2.3 Structure and transition probabilities for stroke sub-model

The structure of the stroke sub-model accommodates both ischemic and hemorrhagic stroke. In addition, to model complications after AIS, this model allows patients to transit to two different survival states in a year after AIS: 1) history of AIS with minor disability due to temporary (no neurologic sequelae) or mild ischemic stroke; 2) history of AIS with major disability due to moderate or severe ischemic stroke. Repeat AIS event enabled from both states with history of AIS.

Figure 7. Structure of Cerebrovascular Sub-model



Abbreviations: IAS, ischemic acute stroke; HS, hemorrhagic stroke.
[1]Minor disability was due to temporary (no neurologic sequelae) or mild ischemic stroke.
[2]Major disability was due to moderate or severe ischemic stroke.

Table 9. Transition probabilities in cerebrovascular disease sub-model (Figure 7)		
Transition	Transition probability	Reference and additional information
A: No stroke to AIS	Stroke equation derived from CRIC study (See Table 10) adjusting the hazard by a factor 1.1928×92.6%.	Clarke et al., 2004; Eriksson et al., 2008
B: No stroke to HS	Stroke equation derived from CRIC study (See Table 10) adjusting the hazard by a factor 1.1928×7.4%.	
C: AIS to history of IS with minor or no disability	22.5%	Piriyawat et al, 2002; Smith et al., 2004; del Zoppo et al., 2009; Spector et al., 1998; Reeves et al., 2010
D: AIS to history of IS with major disability	47.1%	
E: AIS to stroke death	30.4%	
F: Hx of AIS and minor or no disability to AIS (repeat stroke)	3.0%	Derived from data collected from BASIC study (Skolarus et al., 2013); details not included.
G: Hx of AIS and major disability to AIS (repeat stroke)	3.0%	
J: HS to stroke death	53%	Arbois et al, 2000
K: HS to Hx of HS	47%	
H: Repeat AIS to history of IS with minor or no disability	13.2% if a subject had minor or no disability before repeat AIS; 0% if a subject had major disability before repeat AIS	Derived from data collected from BASIC4 study (Skolarus et al., 2013) and Hardie et al. (2004).
I: Repeat AIS to history of IS with major disability	28.0% if a subject had minor or no disability before repeat AIS; 39.2% if a subject had major disability before repeat AIS	
L: Repeat AIS to death	58.8% if a subject had minor or no disability before repeat AIS; 60.8% if a subject had major disability before repeat AIS	

Table 10. CRIC Prediction models				
Parameter	ESKD	MI	CHF	Stroke
$\lambda^{\S}$	0.0052	0.00070	0.0196	0.0382
$\rho^{\S}$	1.022	1.928	0.994	0.44
	$\beta^{\S}$ (SE)	$\beta^{\S}$ (SE)	$\beta^{\S}$ (SE)	$\beta^{\S}$ (SE)
Age at diabetes diagnosis†	-0.019 (0.007)	0.030 (0.012)		
Sex (Female vs. Male)	-0.268 (0.106)			
Hemoglobin A1c (per %)†		0.110 (0.053)		
BMI (per kg/m^2)*	-0.038 (0.010)			
$(\text{BMI} - 40)_+^{\ddagger}$	0.058 (0.022)			
BMI^{2†}				
SBP (per mmHg)†	0.0045 (0.0020)	0.011 (0.004)	0.0083 (0.0032)	
$(\text{SBP} - 140)_+^{\#}$				0.0169 (0.006)
DBP (per mmHg)†		-0.017 (0.008)	-0.017 (0.006)	
HDL (per mg/dL)†		-0.018 (0.008)		
eGFR (per mL/min/1.73m^2)†	-0.118 (0.006)		-0.018 (0.006)	
Log (e-base) UPCR†	0.461 (0.044)	0.265 (0.061)	0.341 (0.050)	0.339 (0.084)
History of CHF	0.524 (0.110)			0.645 (0.250)
History of MI	0.220 (0.111)	0.420 (0.168)	0.838 (0.135)	
History of stroke				1.404 (0.256)
ESKD treatment (dialysis or transplant)		0.531 (0.200)	-0.525 (0.195)	
Smoking status				0.768 (0.305)
Apparent C-index (SE)	0.904 (0.008)	0.709 (0.021)	0.723 (0.018)	0.762 (0.031)

§Hazard at time t can be calculated as $h(t)= \lambda\rho t^{\rho-1}\exp(\mathbf{X}\beta)$, where $\mathbf{X}\beta$ is the corresponding risk score based on predictors and risk coefficients; t is duration of diabetes.

†Centered variables. Details are included in Table 8.

‡ $(\text{BMI} - 40)_+$= BMI-40 if BMI >40; = 0 otherwise.

$(\text{SBP} - 140)_+$= SBP-140 if SBP >140; = 0 otherwise

5.2.4 Structure and transition probability for other death sub-model

The other death sub-model models the risk of death before receiving kidney dialysis and transplant that not directly related to MI event or stroke event. It is a modified version of the risk equation for other w/o ESKD derived using CRIC study data (See Table 11) (adjusting the hazard by a factor 0.42).

Table 11. CRIC Other Death Prediction Models (Death before ESKD)

Parameter	Age < 60	60 ≤Age<75	Age ≥ 75
	β[§] (SE)	β[§] (SE)	β[§] (SE)
Age at diabetes diagnosis (centered at 80)[††]			0.266 (0.076)
Sex (Female vs. Male)			
Hemoglobin A1c (per %)[†]			
BMI (per kg/m^2)*		-0.0374 (0.0108)	
$(BMI - 40)_+$[‡]			
BMI^2[†]		0.00268 (0.00088)	
SBP (per mmHg)[†]	-0.0156 (0.0070)		
$(SBP - 140)_+$[#]			
DBP (per mmHg)[†]			
HDL (per mg/dL)[†]			
eGFR (per mL/min/1.73m^2)[†]			
Log (e-base) UPCR[†]	0.408 (0.101)	0.158 (0.052)	0.351 (0.105)
History of CHF	0.702 (0.334)	1.14 (0.168)	0.805 (0.28)
History of MI			
History of stroke	0.828 (0.377)	0.661 (0.190)	
ESKD treatment (dialysis or transplant)			
Smoking status			
Apparent C-index (SE)			

[§]Cumulative azard at time t can be calculated as $h(t)= h_0(t)\exp(\mathbf{X\beta})$, where $\mathbf{X\beta}$ is the corresponding risk score based on predictors and risk coefficients; t is duration of diabetes.
For age < 60, $h(t) = 0.00297 \times t^{(0.421)}$;
For 60 ≤Age<75, the cumulative hazard function $H(t)= \exp\{-12.1+5.67 \times \log(t)-0.697 \times [\log(t)]^2\}$
For age ≥75, the cumulative hazard function $H(t)= \exp\{-6.28+2.798 \times \log(t)-0.21 \times [\log(t)]^2\}$

[†]Centered variables. Details are included in Table 12.
[††] Centered at 80 (different from the centered age variable in Table 12).
[‡] $(BMI - 40)_+$= BMI-40 if BMI >40; = 0 otherwise.
$(SBP - 140)_+$= SBP-140 if SBP >140; = 0 otherwise

Table 12. Variable transformation for equations in Table 10 and 11	
Variables	Definition
Hemoglobin A1c (centered)	Hemoglobin A1c -7.5
Age at diagnosis of diabetes (centered)	Age at diagnosis of diabetes -64
BMI (centered)	BMI-35
(BMI – 40)+	=BMI-40 if BMI > 40 = 0 otherwise
BMI squared (centered)	$(BMI-35)^2$
SBP (centered)	SBP-132
(SBP- 140)+	=SBP-140 if SBP > 140 = 0 otherwise
DBP (centered)	DBP-67
HDL (centered)	HDL - 44
eGFR (centered)	eGFR-39
log UPCR (centered)	Log PCR + 1.20

4.3. Severe hypoglycemia event

MMD-CKD 1.0 models rate of severe hypoglycemia event by level of anti-hyperglycemia treatment

Table 13. Parameters in the prediction model for risk of other death in one year		
Anti-hyperglycemia treatment	Incidence rate	Comments
Intensive lifestyle (diet and exercise/weight loss)	None	
Metformin (one OAD/non-insulin med)	None	
Metformin + Sulfonylureas (two OADs/non-insulin meds)	0.004 event per person per year	Zoungas et al. (2010)
Add Basal insulin to OAD/non-insulin med	0.02 event per person per year	1. Event per patient per year, median 0; 4 events in 243 patients (1.7%) (Holman et al., 2007) 2. 0 severe event in LANMET study (Yki-Järvinen et al., 2006) 3. 0.03 event per patient per year (Bretzel et al., 2008)
Intensive insulin therapy	0.12 event per person per year	0.02-0.35 event per patient per year (Zammitt and Frier, 2005)

4.4. Treatment Module

Different from other proposed models, MMD 3.0 allows a user to control risk factor changes for the following seven types of treatments and one behavior change:

- Treatment for hyperglycemia
- Treatment for hypertension
- Treatment for dyslipidemia
- Beta-blocker treatment
- Anti-platelet treatment
- Anti-coagulant treatment
- Smoking cessation

Given the fact that modern medicines have largely decreased the complication rate in type 2 diabetes through management of these risk factors, it is important to explicitly model these management strategies and allow users to modify them to match the specific scenarios that they are simulating. For example, MMD explicitly models the treatment regimen for hyperglycemia through five treatment stages. Figure 10 shows a mock trajectory of A1c over time under this treatment regimen using 7% as the treatment threshold.

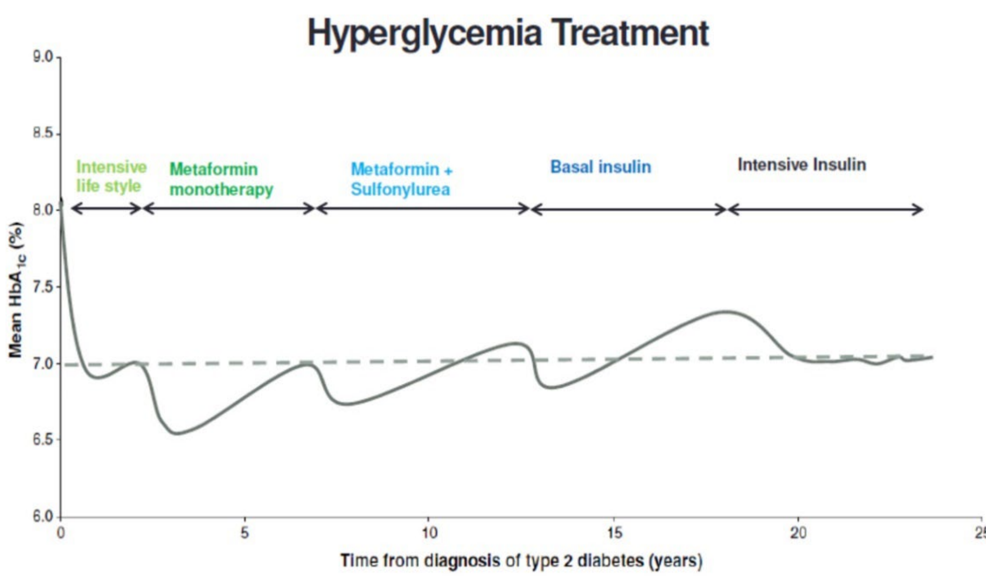


Figure 10. Mock individual trajectory of A1c over time under MMD antiglycemia treatment regimen using 7% as the treatment threshold.

4.4.1 Yearly evaluation and treatment updating rules

The treatment regimens in MMD-CKD 1.0 modifies were developed based on guidelines recommendations by the American Diabetes Association (ADA), American Heart Association (AHA), and American Stroke Association (ASA).

At the beginning of reach simulation interval (year), the model algorithm first defines whether a patient needs initiation of enhancement of certain treatment based on levels of risk factors, disease history or diagnosis, the maximum level of treatment available, and then make the change according to individual adherence characteristics. Table 12 describes the treatment updating rules that define whether an individual needs an update. User can adjust adherence to treatment or annual probability of smoking cessation to model actual update of these rules (see section 4.4.2).

Table 14. Treatment updating rules

Treatment/Medication	Updated Rules	References
Anti-diabetes treatment	For people aged<45, treatment level+1 if A1c>=7% For people aged >=45 and < 64, treatment level+1 if A1c>=7.5% For people aged >=65, treatment level+1 if A1c>=8%	American Diabetes Association. Glycemic Target, 2022; Fox et al., 2015; Inzucchi et al, 2015; Ismail-Beigi et al, 2011
Anti-hypertensive treatment	For people whose SBP>=130 mmHg or DBP>=80 mmHg, treatment level+1	American Diabetes Association. Cardiovascular disease and risk management. 2022; Meschia et al., 2014
ACE-I/ARB	Assume anyone who needs additional anti-hypertensive drug is getting ACE-I/ARB.	
Anti-dyslipidemia treatment (Statin)	For people whose LDL Cholesterol>=150mg/dl, treatment level+1.	American Diabetes Association. Cardiovascular disease and risk management. 2022;
β-blocker	β-blocker should be used in patients with Heart failure with reduced ejection fraction or after MI in patients with preserved left ventricular function	American Diabetes Association. Cardiovascular disease and risk management. 2022;
Warfarin/NOAC	If has atrial fibrillation/atrial flutter and CHADS 2 Score>=2	Meschia et al., 2014
Aspirin/Clopidogrel	For people whose ten year ASCVD score>=0.1, start aspirin	American Diabetes Association. Cardiovascular disease and risk management. 2022;
Non-smoking	Advise all patients not to use cigarettes and other tobacco products or e-cigarettes.	American Diabetes Association. Foundations of care, 2022

4.4.2 Adherence

4.4.2.1 Adherence to medication treatment

Not all patients adhere to recommended treatments. One review found that adherence to oral antidiabetic agents ranged from 36 to 93% across studies and that adherence to insulin was ~63% (Cramer, 2004). In MMD-CKD 1.0, the user can adjust adherence rate for each treatment (i.e. adherence to treatment for hyperglycemia, treatment for hypertension, treatment for dyslipidemia, beta-blocker treatment, anti-platelet treatment, anti-coagulant treatment) to adjust risk factor changes over time.

For simplicity, we assume adherence to each type of treatment is a fixed characteristic of individual patients and does not change before cardiovascular events happen. For each treatment, user defines a rate of adherence in the simulation population. The program then randomly assigns a simulated patient to be adherent or non-adherent based on this population adherence rate.

In addition, we assume when an MI, CHF, or stroke event happens to a patient, he/she might change to adhere to all treatments he/she does not adhere to before the events happen. This adherence parameter is a fixed characteristic of an individual as well.

To implement the above treatment and compliance rules, the simulation program does the following. Before the simulation starts, each patient is assigned a treatment-specific compliance profile that includes seven variables: one for adherence with CVD history and six for treatment-specific adherence rates.

We simulate these adherence variables in a hierarchical structure. For ease of exposition, let's assume 90% of patients comply with all treatments when there is a CVD event, 80%, 70%, 60%, 50%, and 40%, 30% adhere to treatment for hyperglycemia, beta-blocker, dyslipidemia, hypertension, and anti-platelets, and anti-coagulants, respectively. This means 90% of patients will be adherent to all treatments after experiencing a MI, stroke, or CHF event. Among the above 90% of patients, 8 out of 9 (80% of the initial sample) adhere to treatment for hyperglycemia regardless of their CVD complication history; among the 80% of adherent patients with treatment for hyperglycemia, 7 out of 8 (70% of the initial sample) adheres to beta-blocker treatment, etc.; among the total population, 30% adheres to all six treatments regardless of their CVD complication history.

When a patient who is on ACEI/ARB develops ESKD (under dialysis or received transplant) or has eGFR<15 mL/min/1.73m^2, he/she stops taking ACEI/ARB with a 40% rate.

5.4.3.1 Adherence to smoking cessation

All smokers quit smoking each year at a user-specified yearly rate.

4.4.3 Treatment effects on risk factors

At the beginning of each simulation cycle (year), after the algorithm decides treatment changes based on treatment rules and an individual's adherence characteristics, the program calculates the immediate change in weight, HbA1c, lipids and blood pressures, and update these risk factors accordingly.

5.4.3.1 Antiglycemic treatment and weight and HbA1c

MMD 3.0 models six level of antiglycemic treatments

0: No treatment
1: Diet and exercise
2: Oral antidiabetic drug (OAD) /non-insulin medication (metformin)
3: Two oral/non-insulin medications (metformin + sulfonylureas)
4: Basal insulin + OAD
5: Intensive bolus insulin

Change of antiglycemic treatment directly affects weight and HbA1c level in MMD-CKD 1.0.

Table 15. Changes of body weight under different anti-hyperglycemia treatment		
Anti-hyperglycemia treatment	Initial effect (first year change)*	Comments
No treatment → Intensive lifestyle	Mean change*= -3.7kg SD of change=3.5kg	Baseline 80.4kg (SD 15.6 kg) UKPDS 13 (1995)
Intensive lifestyle → one OAD	Mean change= -2kg SD of change=0.3kg	Kahn et al. (2006)
One OAD → two OADs	Mean change 2kg SD of change=1kg	Phung et al. (2010)
two OADs → basal insulin + OAD	Mean change= 1.9kg SD of change=4.2kg	Holman et al. (2009)
basal insulin + OAD → Intensive insulin therapy	Mean change= 1.2kg SD of change=0.5kg	Rosenstock et al. (2009)

Table 16. Changes of HbA1c under different anti-hyperglycemia treatment scenarios		
Anti-hyperglycemia treatment	Initial effect (first year change)	Comments
No treatment → Intensive lifestyle	Mean change=-1.9%-0.5*(currentHbA1c-9.1%) SD of change=abs(mean change)/3	UKPDS 13 (1995)
Intensive lifestyle → one OAD	Mean change=-1.0%-0.5*(currentHbA1c-8.3%) SD of change=abs(mean change)/3	Sherifali et al. (2010)
One OAD → two OADs	Mean change=-0.8%-0.5*(currentHbA1c-8.3%) SD of change=abs(mean change)/3	Phung et al. (2010)
two OADs → basal insulin + OAD	Mean change=-0.8%-0.5*(currentHbA1c-8.4%) SD of change=abs(mean change)/3	Holman et al. (2007)
basal insulin + OAD → Intensive insulin therapy	Mean change = -1.2-0.5*(currentHbA1c-8.4%)	Holman et al. (2009)

	SD of change=0.16	

When a patient newly adapts to intensive lifestyle treatment, the model also applies 6 mmHg and 3 mmHg decrease in SBP and DBP, respectively.

4.4.3.1 Anti-dyslipidemia treatment and lipids level

MMD-CKD 1.0 models three level of anti-dyslipidemia treatments

0: No treatment
1: low dose statin Moderate potency statin
2: high dose statin High potency statin

For each of these two levels, the drug-induced change is 25% decrease, 5% increase, and 6% decrease in LDL-C, HDL-C, and triglyceride, respectively.

4.4.3.2 Anti-hypertensive treatment, beta-blocker treatment and blood pressure level

MMD 3.0 models seven level of anti-hypertensive treatments.

1. No treatment
2. One medication with half dose
3. One medication with full dose
4. Two medications, including 1st medication with full dose and 2nd medication with half dose
5. Two medications, including 1st medication with full dose and 2nd medication with full dose
6. Three medications, including 1st medication with full dose, 2nd medication with full dose, and 3rd medication with half dose
7. Three medications, including 1st medication with full dose, 2nd medication with full dose, and 3rd medication with full dose

β-blocker treatment is only triggered by the event of MI, revascularization procedures, CHF, or stroke, but not by blood pressures higher than treatment threshold. Although it does decrease BPs, it is NOT counted for the seven level of anti-hypertensive treatments. We assume ACE-I/ARB is always the first drug to be added regardless of whether a patient is receiving β-blocker or not.

Table 17. Effect of beta-blocker and anti-hypertensive treatment

<table>
<tr><td colspan="2">Non-β-blocker anti-hypertensive treatment effect</td></tr>
<tr><td>Anti-hypertensive treatment change</td><td>Drug effect</td></tr>
<tr><td>No drug → one drug half standard dose</td><td>If patient is not already on β-blocker
Median change of SBP in mmHg $=-6.9-0.08(SBP-150)$
Median change of DBP in mmhg $=-3.7-0.09(DBP-90)$

If patient is already on β-blocker
Median change of SBP in mmHg $=-3.4- 0.04(SBP-150)$
Median change of DBP in mmHg $=-1.8- 0.04(DBP-90)$</td></tr>
<tr><td>Already on drug → receive an increase of treatment of n levels (n=1,2)</td><td>Median change of SBP in mmHg $=-n\times3.4- n\times0.04(SBP-150)$
Median change of DBP in mmHg $=- n\times1.8- n\times0.04(DBP-90)$</td></tr>
<tr><td>No drug → β-blocker + anti-hypertensive drug (default ACE-I/ARB</td><td>Median change of SBP in mmHg $=-7.4 -3.4-(0.08+0.04)\times(SBP-150)$
Median change of DBP in mmHg $=-5.6-1.8-(0.09+0.04)\times((DBP-90)$</td></tr>
<tr><td colspan="2">β-blocker anti-hypertensive effect</td></tr>
<tr><td colspan="2">If the patient is not other anti-hypertensive treatment
Median change of SBP in mmHg $=-7.4 -0.08(SBP-150)$
Median change of DBP in mmHg $=-5.6-0.09(DBP-90)$

If the patient is already on other anti-hypertensive treatment
Median change of SBP in mmHg $=-3.4- 0.04(SBP-150)$
Median change of DBP in mmHg $=-1.8- 0.04(DBP-90)$</td></tr>
</table>

4.4.4 Treatment direct effect on risk of complications

Table 18. Medication direct effect on risk of complications (not mediated through risk factors)			
Treatment/Medication	Direct risk reduction (hazard ratio)		
	MI event risk	CHF event risk	Ischemic stroke event risk
Statin	0.66 (Freemantle et al., 1999)		
ACE-I/ARB	0.8 (Hope Study Investigators, 2022)	0.75 (Cheng et al., 2014)	
Warfarin			0.32 (Kahwati et al., 2022)
NOAC			0.256 (Kahwati et al., 2022; Patti et al., 2017)
Aspirin/Clopidogrel	0.8 for Male (Pignone et al., 2010)		0.88 (ASCEND Study Collaborative Group, 2018)
β-blocker	0.8 for subjects who are 70 years of age or older or African American, and 0.7 otherwise (Freemantle et al., 1999; Cheng et al., 2014)	0.8 (Garcia-Egido et al., 2015)	

When multiple medications with direct risk reduction are taken by an individual, the maximum of risk reduction among the medications are used. For example, for an individual who is taking statin and ACE-I/ARB at the same time, 34% risk reduction is used for MI event.

4.5 Risk Factors Progression Module

MMD-CKD 1.0 also models diabetes progression through risk factor changes over time. After simulating complication events, the model will generate changes for each of the risk factor from the start to the end of the simulation year and apply the change to calculate updated risk factor levels.

4.5.1. Change in body weight

Table 19. Yearly change of body weight under different anti-hyperglycemia treatment		
Anti-hyperglycemia treatment		**Comments**
No treatment	Mean change=0.8-1kg/year SD of change=0.3kg/year	
Intensive lifestyle	Mean change=1kg-1/year SD of change=0.3kg/year	UKPDS 13 (1995)
one OAD	Mean change=-0.3kg-1/year SD of change=0.3kg/year	Kahn et al. (2006)
two OADs	Mean change=0-1 kg/year SD of change=0.3 kg/year	Phung et al. (2010)
basal insulin + OAD	Mean change=0.8-1kg/year SD of change = 0.5kg	Holman et al. (2009)
Intensive insulin therapy	Mean change=0.8-1kg/year SD of change=0.5kg/year	Rosenstock et al. (2009)

4.5.2. Change in HbA1c

Table 20. Yearly change of HbA1c under different anti-hyperglycemia treatment scenarios		
Anti-hyperglycemia treatment	Changes after one year	Comments
No treatment	Mean change=0.375%/year SD of change=abs(mean change)/3	UKPDS Group (1998) Figure 2 showed 1.5% increase in 6 years. It was arbitrarily increased to reflect faster increase without any treatment. An arbitrary variation was added to allow the change to be between zero and twice the value calculated from the references.
Intensive lifestyle	Mean change=0.25%/year SD of change=2/3	UKPDS 33 (1998)
One OAD	Mean change=0.14%/year SD of change=2/3	Kahn et al. (2006)
Two OADs	Mean change=0.2%/year SD of change=2/3	Charbonnel et al. (2005)
basal insulin + OAD	Mean change=0.2%/year SD of change=2/3	Rhoads et al. (2011)
Intensive Insulin	Mean change = 0%/year SD of change=2/3	Holman et al. (2009)

4.5.3. Change in lipids

Annual changes in lipids are calculated based on three equations derived from secondary analysis using a multivariate model and ARIC, CHS, and CARDIA datasets.

All changes are first calculated for log (e-based) transformed lipid levels based on level of log(HDL), log(LDL), log(triglycerides) at the beginning of the year, change of log (e-base) transformed fasting glucose and change in BMI from the start to the end of the year, age, and sex.

In order for simulate correlated changes in HDL, LDL, and triglycerides, we first simulate three independent standard normal variables, ε_1, ε_2, and ε_3. Then correlated variations in change of the three lipids were added to the calculated mean change levels based on the covariance matrix estimated in the above multivariate model, as shown the following three equations.

Table 21. Equations for calculating change in lipids	
Lipids	Equations
Change in log (HDL)	0.0140 + Age × (-.00112)+Age^2 × 0.0000117+ log(triglycerides) × (-.0145)+log(LDL) × (-.000961)+log(HDL) × (-.0844)+ change in log(fastingGlucose) × (-.0364)+ change in BMI × (-.00414)+Female × (0.0147) + 0.0648 × ε_1
Change in log (LDL)	0.0738 + Age × 0.00412+ Age^2 × (-.0000463)+ log(triglycerides) × (0.0114)+ log(LDL) × (-.138)+ log(HDL) × (0.00620)+ change in log(fastingGlucose) × 0.0821+ change in BMI × 0.00906+Female × 0.00600+0.111 × ε_1+0.00206 × ε_2
Change in log (triglyceride)	-.157 + Age × 0.00728+ Age^2 × (-.0000660) + log(triglycerides) × (-.112) + log(LDL) × 0.0189 + log(HDL) x (-.0496) + change in log(fastingGlucose) × 0.268 + change in BMI *0.0275 + Female*0.0215 + 0.1359 × ε_1+0.00734 × ε_2-0.0189 × ε_3

4.5.4. Change in BPs

Annual changes in SBP and DBP are calculated based on two equations derived from secondary analysis using a multivariate model and ARIC, CHS, and CARDIA datasets.

All changes are first calculated for SBP and DBP levels based on level of SBP and DBP at the beginning of the year, change in BMI from the start to the end of the year, age, and race. In order for simulate correlated changes in SBP and DBP, we first simulate two independent standard normal variables, ε_4 and ε_5. Then correlated variations in change of the two BP levels were added to the calculated mean change levels based on the covariance matrix estimated in the above multivariate model, as shown in the following two equations.

Table 22. Equations for calculating change in BPs	
Lipids	Equations
Change in DBP	-2.6+Age × 0.28+DBP × 0.031+SBP × 0.031+Age × SBP × (-0.00077)+Age × DBP × (-0.0031) + BMI_Diff × 0.37+Female × (-0.38)+AfricanAmerican × 0.57+2.58 × ε_4
Change in SBP	-34.18733961+Age×1.02+DBP×0.132+SBP×0.186+Age×SBP× (-0.0059)+Age×DBP× (-0.0027)+BMI_Diff × 1.79 + Female × 0.53 + AfricanAmerican × 0.97 + 7.3 × ε_4+2.5 × ε_5

4.5.5. Change in UACR/UPCR

We developed our model for yearly change in urine albumin creatinine ratio/urine protein creatinine ratio (UACR/UPCR) using two databases, The Chronic Renal Insufficiency Cohort (CRIC) and Action to Control Cardiovascular Risk in Diabetes (ACCORD). CRIC study enrolled a population with chronic kidney disease (including a subpopulation with type 2 diabetes) and measured UPCR but not UACR at yearly follow-up. It provided a good sample size for modeling changes in UPCR for type 2 diabetes patients with eGFR $\leq$60 mL/min/1.73m^2. ACCORD trial enrolled type 2 diabetes patients with or without kidney complications. Majority of the ACCORD patients had eGFR > 60 mL/min/1.73m^2. ACCORD study measured UACR at 24 months and 48 months follow-up visits. It provided a good sample for modeling changes in UACR among type 2 patients with eGFR > 60 mL/min/1.73m^2.

We used CRIC data to develop a model for yearly change in log UPCR for subjects with eGFR $\leq$ 60 mL/min/1.73m^2 using the following equation.

$$\text{Yearly change in UPCR} = \gamma_0 + \gamma_1 \times \text{log current UPCR} + \gamma_2 \times \text{current eGFR} + \boldsymbol{\delta X} + \varepsilon$$

To calculate yearly change of log UACR for subjects with eGFR > 60 mL/min/1.73m^2, we used ACCORD data to develop to models for UACR $\leq$ 0.3 g/g and UACR > 0.3 g/g, respectively using similar equations.

$$\text{Yearly change of UACR} = \gamma_0 + \gamma_1 \times \text{log current UACR} + \gamma_2 \times \text{current eGFR} + \boldsymbol{\delta X} + \varepsilon$$

To develop each of these models, candidate predictors included were sex, race and ethnicity, current BMI, current HbA1c, current SBP, prior MI or revascularization, prior CHF, smoking status, age at first diabetes, current age, yearly change in SBP, yearly change in BMI, yearly change in HbA1c, and interaction between current HbA1c and current log UACR.
We used splines to explore the functional forms of all risk factors considered and backward selection with exclusion criteria p>0.05 to find the best prediction models. These models allowed us to capture the heterogeneous individual trajectories of these risk factors over time. In addition, we developed an equation for conversion between UACR and UPCR using baseline data in the CRIC study.

For updating UACR and UPCR in MMD-CKD, we first calculate the changes in log (UACR) or log (UPCR) using the corresponding equations in Table 22. When an equation for calculating change in log (UACR) is used, we consequently update UPCR by converting the updated UACR level [UPCR (g/g)=exp(0.43388 + 0.72396 $\times$ log (UACR (g/g)))] . If change in log (UPCR) is used, we consequently update UACR by converting the updated UPCR level [UACR (g/g)=exp(-0.5993+1.38129$\times$log (UPCR(g/g)))]. This converting equation was based on linear regression using baseline CRIC data (more details can be found in Ye et al. 2022) .

Table 22. Equations for calculating UACR/UPCR changes.				
Current eGFR	Current UACR	Equation for calculating change in UACR or UPCR	Data	Comment

eGFR > 60 mL/min/1.73 m^2	UACR $\leq$ 0.3 g/g	ΔLog(UACR)= -0.868655+0.12+0.090549*(gender=="Male")+ 0.087299*(Hba1c-8)*(HbA1c>=8) + 0.005198*SBP+ 0.021439*(SBP_Diff)+ 0.137308*(HbA1c_Diff)-0.003493*(eGFR)-0.136403*log(uacr_gperg)+ 0.58*Min(3, Max(-3, Gaussian(0,1)))	ACCORD	UACR unit is g/g
eGFR >60 mL/min/1.73 m^2	UACR > 0.3 g/g	ΔLog(UACR- -0.868655+0.12+(0.090549+0.39087)*(gender=="Male")-0.300439+ (0.087299+0.084308)*(HbA1c-8)*(HbA1c>=8) + (0.005198-0.003539)*SBP + (0.021439+0.001931)*(SBP_Diff)+ (0.137308-0.008869)*(HbA1c_Diff)+(-0.003494+0.009808)*eGFR+(-0.136403+0.353556)*log(uacr_gperg) + 0.58*Min(3, Max(-3, Gaussian(0,1)))	ACCORD	UACR unit is g/g
eGFR $\leq$60 mL/min/1.73 m^2		ΔLog(UPCR)=-0.2682-0.0756*(1-Male)+0.0537*Iif(Ge(HbA1c-8, 0), HbA1c-8, 0)+0.0056*SBP-0.0942*logUPCR-0.0696*logUPCR*logUPCR-0.0272*logUPCR*logUPCR*logUPCR-0.0082*eGFR-0.0030*AgeAtDiagnosisOfDiabetes+0.0108*SBP_Diff+0.07358*HbA1c_Diff+0.72* Min(3, Max(-3, Gaussian(0,1)))	CRIC	UPCR unit is g/g

4.5.6 Change in estimated glomerular filtration rate

Similar to the equations for updating UACR and UPCR, we also used CRIC for modeling yearly changes in eGFR among those with eGFR ≤60 mL/min/1.73m^2, and used ACCORD data for modeling yearly changes in eGFR among those with eGFR > 60 mL/min/1.73m^2.

To calculate yearly change by eGFR for subjects with current eGFR ≤ 60 mL/min/1.73m^2, we developed three models based on CRIC data for current UPCR < 0.15g/g, 0.15 ≤ UPCR < 0.50 g/g and UPCR ≥ 0.50g/g, using the following equation.

$$\text{Yearly change of eGFR} = \gamma_0 + \gamma_1 \times \text{log current UPCR} + \gamma_2 \times \text{current eGFR} + \delta\mathbf{X} + \varepsilon$$

where **X** is a set of other risk factors (e.g., age, duration of diabetes, systolic blood pressure, HbA1c, change in BMI, etc.) selected to predict the yearly change in eGRF and ε is random error following a normal distribution.

To calculate yearly changes in eGFR for subjects with current eGFR > 60 mL/min/1.73m^2, we developed two models based on ACCORD data for current UACR <0.30g/g and UACR ≥ 0.30g/g, respectively, using a similar equation.

$$\text{Yearly change of eGFR} = \gamma_0 + \gamma_1 \times \text{current log UACR} + \gamma_2 \times \text{current eGFR} + \delta\mathbf{X} + \varepsilon$$

Same methods described in 4.5.4 were used for developing these models and the results were shown in Table 24.

Table 24. Equations for calculating eGFR changes.		

Current eGFR	Current UACR	Equation for calculating change in UACR or UPCR	Data	Comment
eGFR > 60 mL/min/1.73m^2	UACR ≤ 0.3 g/g	ΔeGFR =- 13.28825-0.8633*age-0.23195*HbA1c-0.11667*eGFR-0.6378*log(uacr_gperg)+0.0609*(SBP_Diff)-1.17886*(BMI_Diff)+2.2367*(log(uacr_gperg)_Diff)+8.2*Min (3, Max(-3, Gaussian(0,1)))	ACCORD	UACR unit is g/g
eGFR > 60 mL/min/1.73m^2	UACR > 0.3 g/g	ΔeGFR=- 3.0721349-3.2582582*log(uacr_gperg)-0.10236123*eGFR+0.0007006*(SBP_Diff)-0.3314607*(BMI_Diff)+7.8*Min(3, Max(-3, Gaussian(0,1)))	ACCORD	UACR unit is g/g
eGFR ≤60 mL/min/1.73m^2	UPCR < 0.15 g/g	ΔeGFR=0.966-0.107*Current eGFR-1.083*Current log UPCR+ 0.09269*Change in SBP+0.83546*Change in log UPCR+Min(3, Max(-3, Gaussian (0,1)))*sqrt(46.2)	CRIC	UPCR unit is g/g
eGFR ≤60 mL/min/1.73m^2	0.15 <=UPCR < 0.5 g/g	ΔeGFR=2.375-0.124*Current eGFR-0.871*Current log UPCR+ 0.087*Change in SBP-0.384*Change in BMI+1.133*Change in log UPCR+ Min(3, Max(-3, Gaussian (0,1)))*sqrt(45.7)	CRIC	UPCR unit is g/g
eGFR ≤60 mL/min/1.73m^2	UPCR >=0.5 g/g	ΔeGFR=-7.507-0.100*Current eGFR-0.023*Current SBP-1.999*Current log UPCR+0.598* Current BMI-0.0081*Current BMI^2 +0.062*Change in SBP+ Min(3, Max(-3, Gaussian (0,1)))*sqrt(40.7)	CRIC	UPCR unit is g/g

4.6 Costs Module

Table 25. Costs of complications for Michigan Model for Diabetes

Event and ongoing costs of complications for Michigan Model for Diabetes	2020 US dollars[b]		Sources
	Event	Ongoing	
Baseline cost[a]		2,633	Brandle et al., 2023
Nephropathy[c]			
Chronic kidney disease – stages 1 or 2	0	0	Lage et al. 2019
Chronic kidney disease – stages 3a or 3b	387	387	
Chronic kidney disease – stages 4 or 5	1,766 per one unit reduction (mL/min/1.73m^2) in eGFR	1,766 per one unit reduction (mL/min/1.73m^2) in eGFR	
End-stage renal disease with hemodialysis	112,656	112,656	U.S. Renal Data System, USRDS 2013 Annual Data Report
End-stage renal disease with renal transplant	157,043	50,423	
Coronary heart disease			
Angina	9,420	2,433	O'Brian et al., 2000
Myocardial infarction	47,480	2,624	
Percutaneous transluminal coronary angioplasty[c]	9,420	2,433	
Coronary artery bypass graft[c]	69,024	2,644	
Myocardial infarction with coronary artery bypass graft[c]	69,024	2,644	
Congestive heart failure	39,394	8,677	Liao et al., 2006
Cerebrovascular disease			
Ischemic stroke	65,781		O'Brian et al., 2000
With no/minor disability after ischemic stroke		12,270	O'Brian et al., 2000; O'Brian and Gage, 2005; Freeman et al., 2011; Harrington et al., 2013; Nguyen et al., 2016
With major disability after ischemic stroke		31,636	
Hemorrhagic stroke	182,823	40,779	
Hypoglycemia requiring hospitalization	19,325		Ward et al., 2014
Death, by age in years[c]			
74 or younger	85,051	NA	
75-84	68,915	NA	
85 or older	46,811	NA	

Abbreviations: eGFR, estimated glomerular filtration rate; NA, not applicable.

[a]The baseline cost is the annual direct medical cost for a white man with type 2 diabetes and BMI of 30 kg/m^2 who is treated with diet and exercise and has no microvascular, neuropathic, or cardiovascular complications.

[b]Costs are expressed in year 2020 US dollars using the Personal Consumption Expenditures-Health price index.

[c]Staging of chronic kidney disease (CKD) is defined by the eGFR levels, including stage 1 CKD (eGFR ≥90), stage 2 CKD (eGFR: 60-89), stage 3a CKD (eGFR: 45-59), stage 3b CKD (eGFR: 30-44), stage 4 CKD (eGFR: 15-29), and stage 5 CKD (eGFR <15). The event and ongoing costs for stages 1-5 CKD include the incremental CKD-related non-drug costs, i.e. CKD-related outpatient, acute care (hospital and emergency room), and dialysis costs.

[d]According to the statements in 2 JACC papers, about one third of patients undergoing PCI in the US have diabetes (see page e83 in the attached File 1) and about 35% of CABG patients have diabetes (see page e167 in the attached File 2). Also, according to a recent Circulation paper, it was estimated that in 2010, in the US, 492,000 patients underwent PCI while 219,000 underwent CABG (see page e275 in the attached File 3). With calculations using these data, what we could have is: The estimated number of diabetic patients treated with PCI in 2010 in the US would be 164,000 (=492,000*1/3), while that treated with CABG would be 76,650 (=219,000*0.35). Thus, based on these 2

calculated numbers, we could get that about 68% of diabetic patients who need the coronary revascularization procedures may use PCI, while 32% of them may get CABG.

Table 26. Costs of Medications for Michigan Model for Diabetes (details of deriving these numbers are omitted).

	2020 US dollars	Comments and sources
Medication cost (per year)		
Warfarin	$562	Including drug and monitoring costs
NOAC	$6539	Micromedex 2.0
Antiplatelet Treatment (Aspirin or clopidogrel)	$164*	Micromedex 2.0
Beta-Blocker	$1372	Micromedex 2.0
Hypertensive treatment		
Level 1: One med with half dose	$26.2	Micromedex 2.0
Level 2: One med with full dose	$34.5	
Level 3: Two meds, including 1st med with full dose and 2nd med with half dose	$73.8	
Level 4: Two meds, including 1st med with full dose and 2nd med with full dose	$52.4	
Level 5: Three meds, including 1st med and 2nd med with full dose, and 3rd med with half dose (per year)	$91.6	
Level 6: Three meds, including 1st med, 2nd med, and 3rd med with full dose	$100	
Treatment for hyperglycemia		
Level 1: Intensive lifestyle (IL)	$1,764	Micromedex 2.0
Level 2: One non-insulin medication	$597	
Level 3: Two non-insulin medications	$1,113	
Level 4: Add basal insulin[1]	$6,550	
Level 5: Intensive insulin therapy (basal/bolus)[2]	$19,621	
Statin		
Moderate potency	$161	Micromedex 2.0
High potency	$161	
*a weighted average based on % of people taking aspirin and clopidogrel		

4.7 Quality of Life Module

Health utility scores are a measure of HRQoL, in which perfect health is assigned a value of 1.0 and death is assigned a value of 0.0. The MMD-CKD incorporates a health utility module to calculate yearly and cumulative QALYs based on subjects' demographic, diabetes treatment, complication, and comorbidity status. In each year, MMD-CKD calculates the yearly utility score for each simulated individual by subtracting penalties related to complications from the baseline utility. For example, for a female with BMI ≥30:

Yearly utility score = 0.689 - 0.038 - 0.021 - penalty related to diabetes intervention - penalty related to nephropathy - penalty related to neuropathy - penalty related to cardiovascular disease - penalty related to cerebrovascular disease - penalty related to high blood pressure

For most of the submodels, the complication levels are exclusive. The only exception is cardiovascular disease, for which we take the largest penalty that applies. For example, for an individual with both CHF and history of MI, we will use -0.052 as the penalty for cardiovascular disease.

We assume each individual's utility score is constant during each year. In economic analyses, the health utility score for each health state is multiplied by the time an individual spends in that health state, in our case, yearly utility score times 1. These are then summed to calculate the quality-adjusted life expectancy (QALE), expressed as quality-adjusted life years (QALYs) accrued over a specified period of time.

Table 27. Penalty functions for QWB-SA health utility scores			
Disease status/characteristics	Complication Level	QWB-SA Penalty	Sources
	Intercept	0.689	Coffey et al., 2002
Sex	Male	(Ref)	
	Female	-0.038	Coffey et al., 2002
BMI (kg/m^2)	Obese (BMI ≥30)	-0.021	
Diabetes Intervention	None or diet only	(Ref)	
	Oral/non-insulin antidiabetic agents	-0.023	Coffey et al., 2002
	Insulin	-0.034	
Nephropathy	No nephropathy	(Ref)	
	Stages 1 or 2 CKD	-0.011	Coffey et al., 2002
	Stage 3 CKD	-0.013	
	Stage 4 CKD	-0.022	
	Stage 5 CKD (without dialysis)	-0.022	
	Stage 5 CKD (ESRD dialysis)	-0.078	Coffey et al., 2002
	Stage 5 CKD (ESRD transplant)	-0.078	
Neuropathy	No neuropathy	(Ref)	
	Clinical neuropathy	-0.065	Coffey et al., 2002
	Amputation	-0.105	
Cardiovascular disease	No CHD	(Ref)	
	Angina	-0.026[†]	Coffey et al., 2002
	MI/PTCA/CABG	-0.026[†]	Coffey et al., 2002; Zhang et al., 2012
	CHF	-0.052	
Cerebrovascular disease	No history of stroke	(Ref)	
	Ischemic stroke (IS) with no/minor disability after IS	-0.01	Coffey et al., 2002; Rumberger et al., 2010
	Ischemic stroke (IS) with major disability after IS[2]	-0.17	
	History of hemorrhagic stroke (HS)[3]	-0.20	O'Brian and Gage 2005; Freeman et al., 2011; Harrington et al., 2013; Nguyen et al., 2016; Samarasekera et al., 2015
High blood pressure	High BP or on BP medication	-0.011	Coffey et al., 2002

[†]Coffey et al. (2002) did not provide a penalty for having history of Angina or MI/PTCA/CABG. In Zhang et al. (2012), the penalty for other heart disease is approximately half of the penalty for CHF. We therefore imputed the penalty for Angina and MI/PTCA/CABG as half of the penalty for CHF.

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